

“The modelling of feedback in star formation simulations”

A Summary
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Content

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Stellar Feedback Physics

Photoionization, Main-sequence
line driven winds, Stellar
evolution, Supernovae, Accretion
heating, Radiation pressure, Jets

02

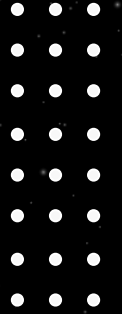


Astrophysical Fluid Dynamics Codes

Grid codes, Particle codes,
Moving mesh codes

Stellar Feedback Physics

Involves the insertion of matter, momentum, and energy from stars in the surrounding fluid.



Photoionization

- The process in which an ion is formed from the interaction of a photon with an atom or molecule
- As the ionization front moves outward, more stellar photons are consumed by recombinations
- Strömgren sphere reached when there is no neutral gas to ionize.
- Temperature of Strömgren sphere determined by an equilibrium of heating and cooling processes
- Effects of photoionization:
 - Supersonic expansion of Strömgren sphere
 - Ionization of the whole cloud
 - Formation of "fingers" of dense neutral material

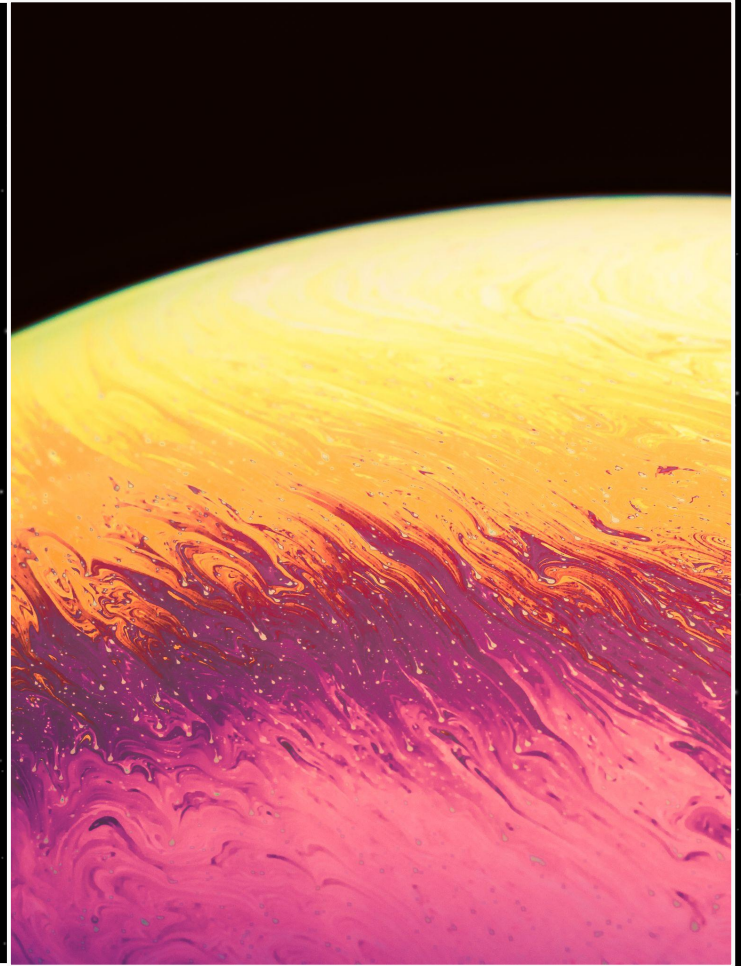
Main Sequence Line-driven Winds

- Powerful fluxes of stars are able to accelerate line-driven winds
- Material leaves the surface of the star at extremely high velocity
- Effect of the wind depends on the thermodynamics of the shocked gas
- Large fractions of the wind energy are lost via leakage of hot wind gas or small-scale mixing with cold gas



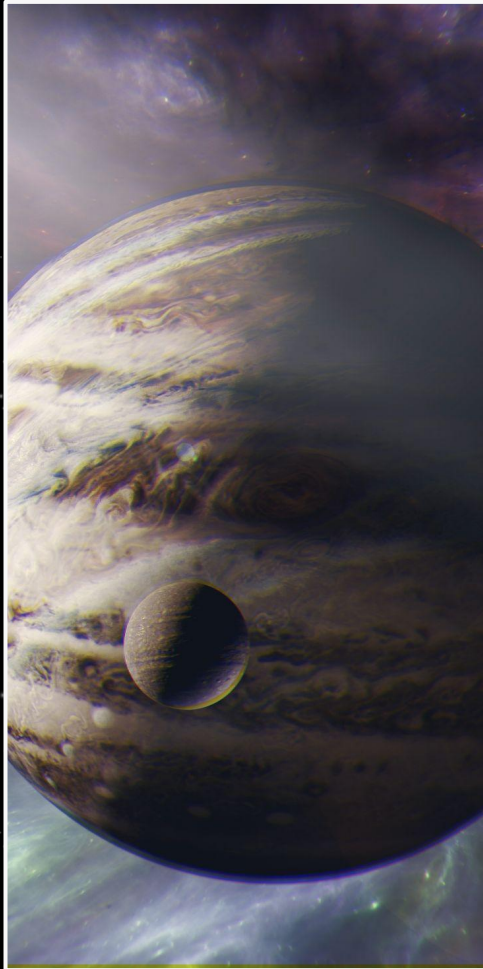
Stellar Evolution

- Lifetimes of massive stars comparable to lifetimes of GMCs
- Some massive stars will enter the final stages of their lives in their natal clouds
- The WR stage profoundly impacts the wind properties
- The mass loss rate increases as the star sheds its outer layers
- Terminal velocity declines
- Impacts photoionization and winds



Supernovae

- When massive stars exhaust their core hydrogen, a chain of events ensues which eventually results in a supernovae explosion
- Ejection of metal at high speeds and energy
- Free-expansion phase
- Sedov-Taylor phase
- Radiative phase



Accretion Heating

- In early stages, protostars are accreting gas from their host GMCs
- When this happens, gravitational potential energy is converted to heat, which the protostar radiates away
- This radiation is absorbed by dust
- If gas density is large enough, there is coupling of the radiation field to the gas

Radiation Pressure

- All stellar photons transfer momentum and thermal energy to the surrounding dust
- The accretion heating process also exerts radiation pressure forces
- Radiation pressure is less important than the thermal pressures unless its from luminous massive stars
- Computing Radiation pressure is difficult, as the momentum transferred needs to be known
- Difficult to compute especially for optically thick gases

Jets

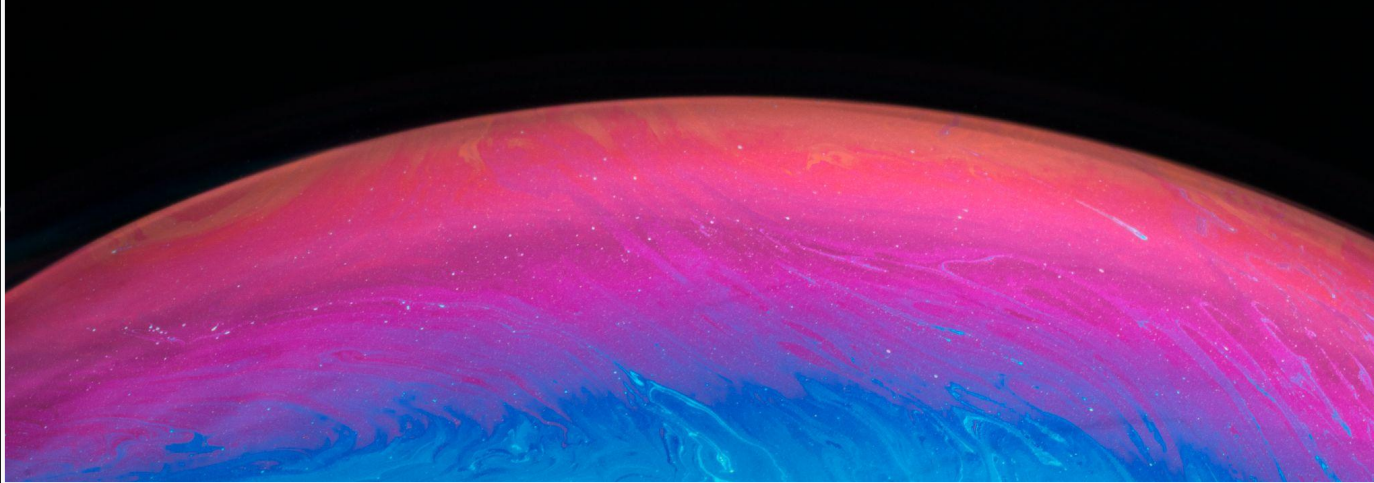
- Accretion also drives the emission of stellar jets
- The origin of jets is from an interaction between its star and its accretion disk
- A magnetic field is produced that acts like a particle accelerator or collimator
- Shocks produced are high radiative, may be considered as sources of momentum only



02

Astrophysical Fluid Dynamics Codes

Interstellar medium is approximated as a smoothly-varying fluid. To model this it is necessary to discretise it into individual fluid elements.



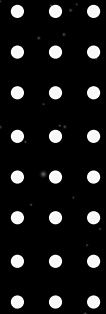
	Grid Codes	Particle Codes	Mesh Codes
Fluid Element	Volume elements as grids with set boundaries	Mass elements as particles (SPH)	Mass elements as particles
Evolved	Forces on fluid calculated and material moved	Forces on each particle and moved	Volume as mesh that follows flow
Advantages	Use any criterion to define grid, wide range of mass contained	Self-computing gravitational forces, zero quantities	Gravitational forces, shocks, redefine resolution
Disadvantages	Boundaries, non-zero quantities of gas, fluid flows, supersonic speed	Shocks, only mass density controls resolution	Relative complexity and novelty
Sink Particles	Decided by density of grid cell, can move and accrete further material	Decided by mass density and treated like gas particles	Implemented like the grid code method

Code Boundaries

When matter crosses a grid boundary, the type of boundary determines the outcome.

- Open/Outflow: matter is destroyed
- Reflective: matter is sent back to a domain as though bouncing off a wall
- Periodic: matter is reinserted at the opposite end of the domain
- Inflow: allows matter to be created and flow into the domain

Some codes allow grid cells to be subdivided or incorporated into larger grids. Codes that do this automatically are Adaptive Mesh Refinement Codes.



THE END!

